

# Global Inequalities in Age-Standardized Alzheimer's Disease Mortality Across Socio-Demographic Index Groups

*Geographic Patterns, Temporal Paradoxes, and Forecasts (1990–2030)*

Shehroz Khan | BSc Public Health  
University of Debrecen  
Supervisors: Dr. Toth Agnes,  
Dr. Ghanem Amr Sayed

# Background: The Diagnostic Divide

## Alzheimer's Disease Global Impact

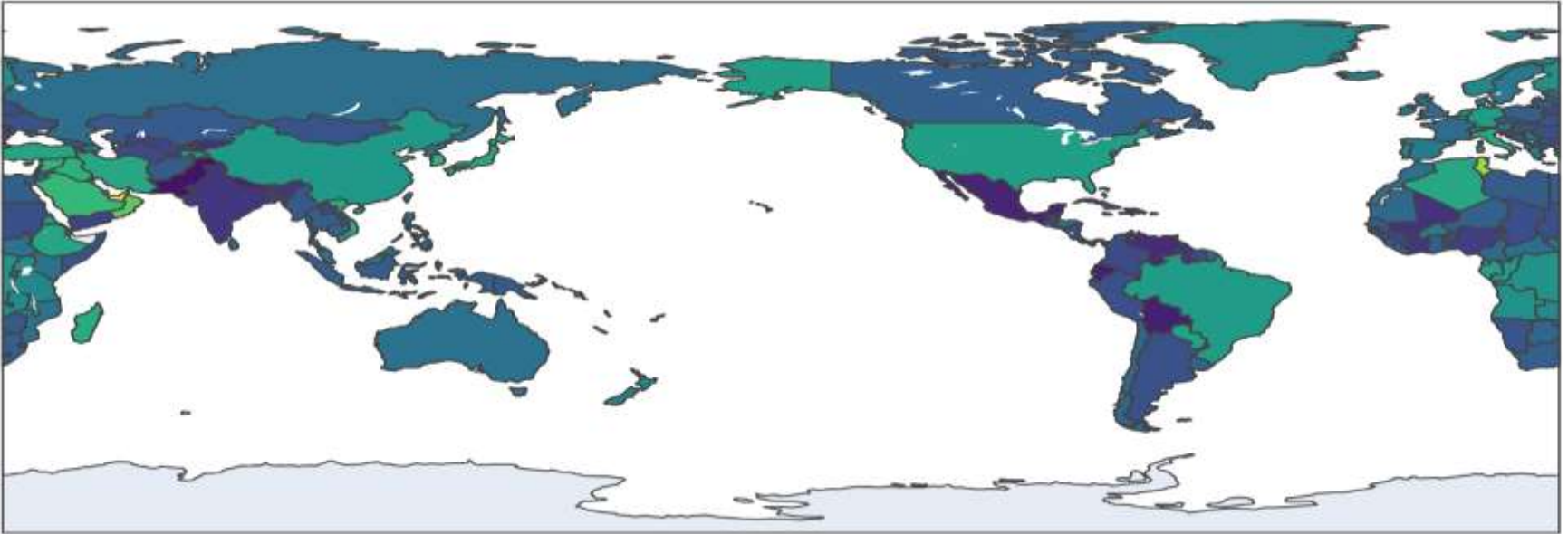
- 55+ million people living with dementia globally
- Projected to reach 139 million by 2050
- 7th leading cause of death worldwide

## The Diagnostic Divide

- **High-income** Alzheimer's ranks #4 cause of death
- **Low-income** Not even in the top 10

**Ranking gap: >10 positions between income groups**

# Global Alzheimer's mortality rate ASMR 2023 per 100,000



- Across-country range (2023): 11.9–43.8 per 100,000



choropleth\_2023.html



# Objectives

1

Quantify SDI–mortality association (global)

2

Evaluate pandemic-period impact via joint point regression

3

Forecast mortality to 2030 with uncertainty intervals

4

Interpret Hungary within Central Europe



# Methodology

## Data & Design

- Data Source GBD 2023
- Outcome: ASMR per 100,000
- Exposure: SDI (continuous) + SDI categories (5 groups)
- Global ecological trend study (GBD 2023; 204 countries; 1990–2023; SDI-stratified; join point + 2030 forecast).

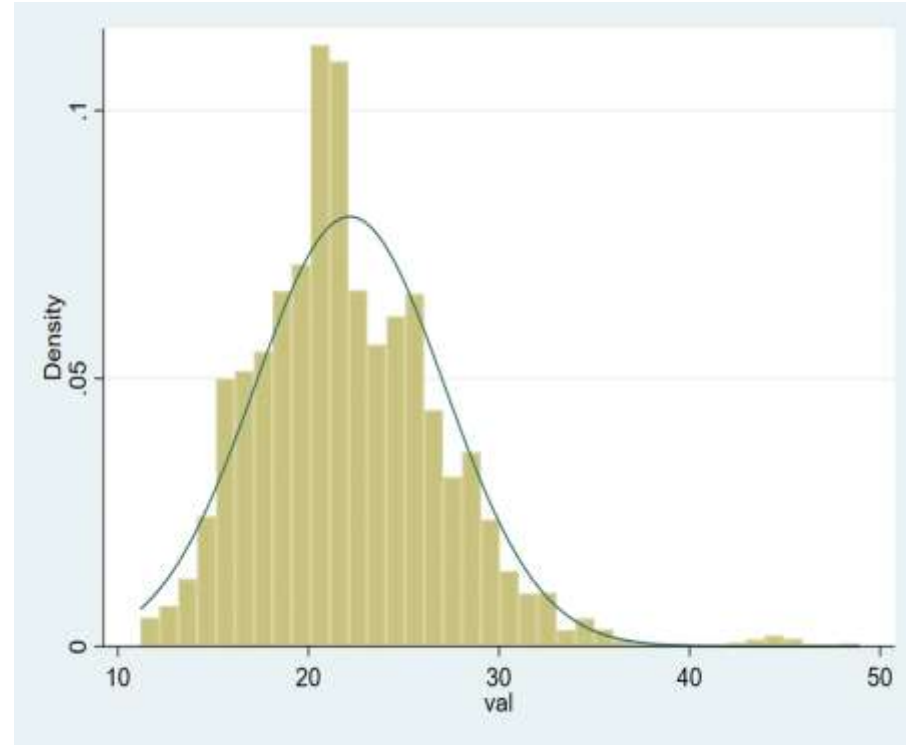
## Statistics

- Shapiro–Wilk test (normality) | Linear regression (SDI → ASMR) | Kruskal–Wallis H test | Dunn’s post-hoc test | Joinpoint regression (trend-change detection) | Time-series analysis | forecasting to 2030 (95% prediction intervals)
- **Software**  
Stata  
Joinpoint

# Normality Test

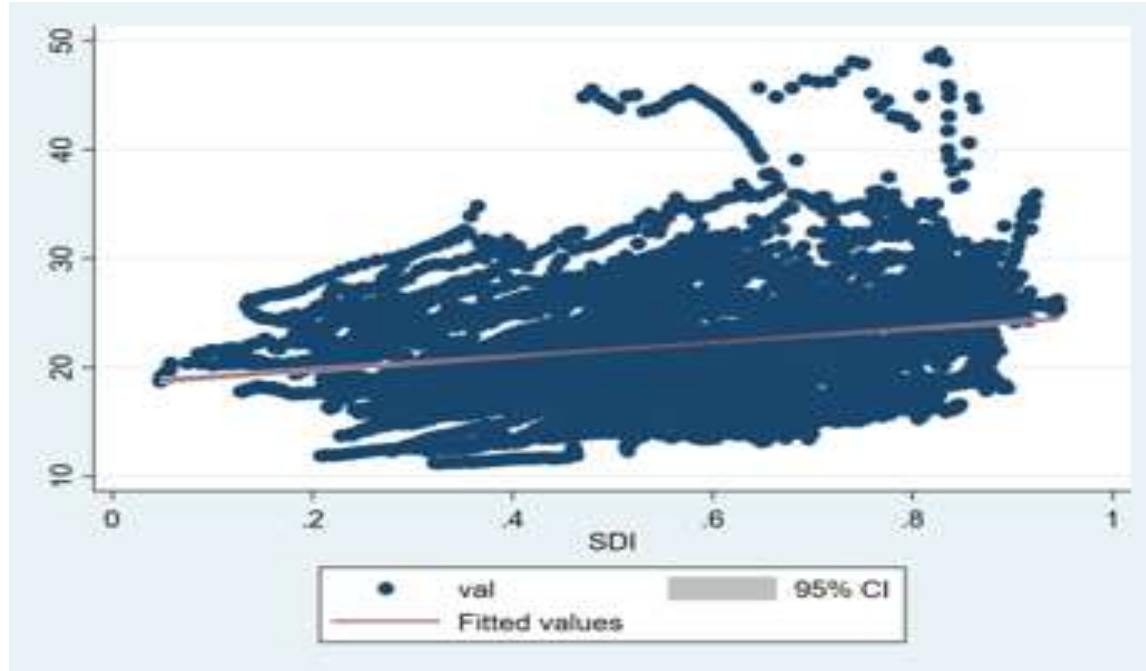
## Shapiro-Wilk Test

$p < 0.001 \rightarrow$  NOT normally distributed.  
Justifies non-parametric tests.





# Linear regression



$\beta(\text{SDI})$

**6.303567**

95% CI

**5.70–6.90**

p-value

**< 0.001**

**N = 6,936 |  $R^2 = 0.0577$**



# Kruskal–Wallis Test and Dunn’s Test

## Kruskal–Wallis Test

$$\chi^2(4) = 682.032, p = 0.0001$$

1 = Low SDI

2 = Low-middle SDI

3 = Middle SDI

4 = High-middle SDI

5 = High SDI

## Group Sizes

SDI1: 1,921 | SDI2: 1,731

SDI3: 1,261 | SDI4: 1,209

SDI5: 814

## Dunn Pairwise

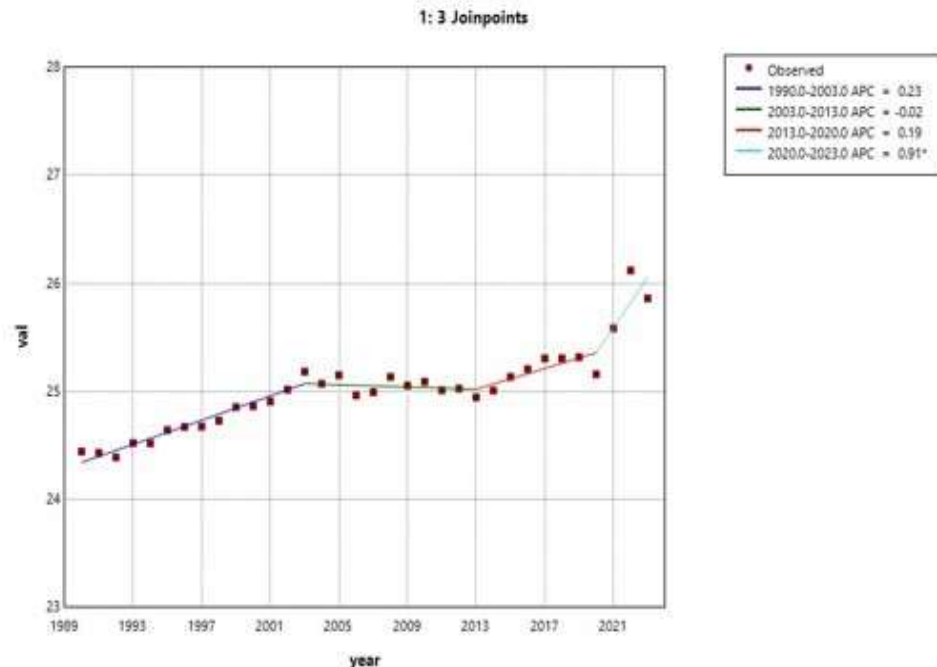
All comparisons except one (Low-middle vs Middle SDI) show significant differences ( $p < 0.001$ )

# The Pandemic Inflection (1990–2023)

## Joinpoint Model

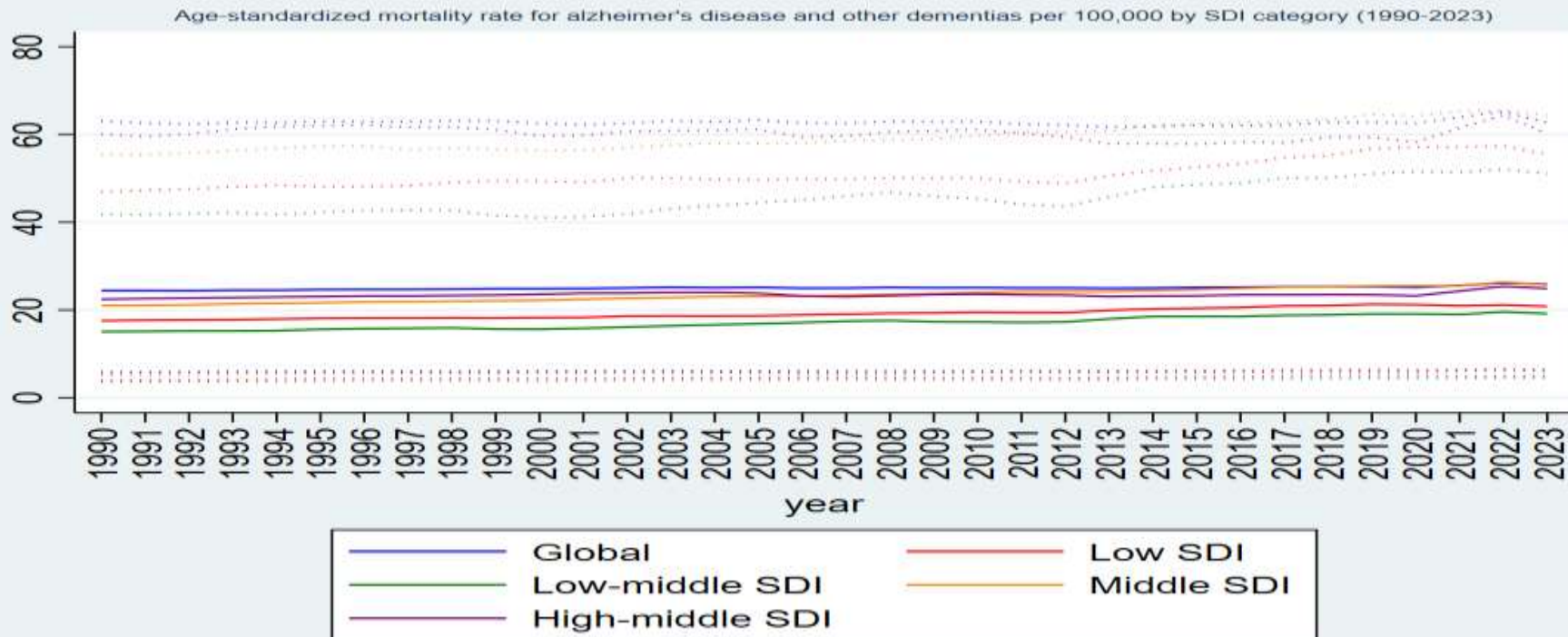
- 1990–2003:  $APC = 0.23$
- 2003–2013:  $APC = -0.02$
- 2013–2020:  $APC = 0.19$

**2020–2023:  $APC = 0.91^*$**   
(significant)



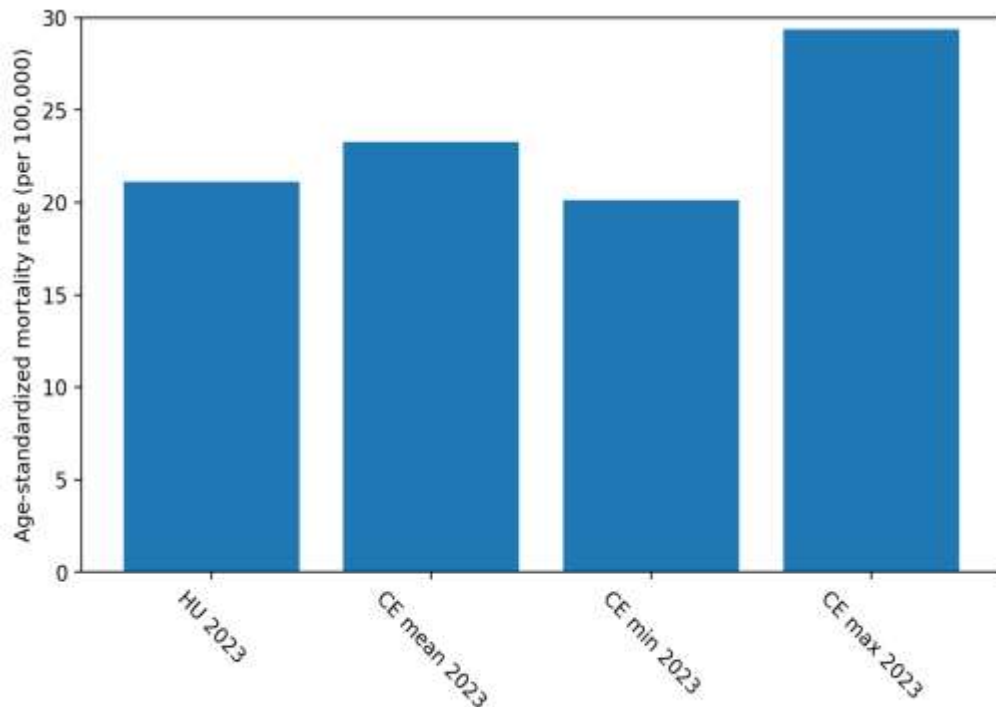
\* Indicates that the Annual Percent Change (APC) is significantly different from zero at the  $\alpha = 0.05$  level.  
Final Selected Model: 3 Joinpoints.

# Time series graph (1990–2023)





# Hungary in Central Europe



Hungary is mid-range in Central Europe (2023).

## Key Statistics (ASMR per 100,000)

Hungary (2023)

**21.1287**

### Long-term stability:

1990:20.7896

2023:21.1287

Change:0.3392 per100,000 (+1.63%)

### Post-2020 change:

20.72 → 21.13 (+1.95%)

### Central Europe's position:

6 / 8 ( highest → lowest)

### Central Europe spread (2023):

20.1129 (Czechia) to 29.3713

(Germany) → range = 9.2584



# Forecast to 2030

**Joinpoint (segmented APC model)**

- Forecast 2030: **25.8 per 100,000**
- 95% CI: **24.2–27.4** (*wider, APC-based*)

| SDI Group       | 2030 ASMR | 95% CI    |
|-----------------|-----------|-----------|
| Global          | 25.8      | 25.4–26.2 |
| Low SDI         | 22.0      | 21.4–22.6 |
| Low-middle SDI  | 20.4      | 19.8–21.0 |
| Middle SDI      | 27.1      | 26.7–27.4 |
| High-middle SDI | 24.3      | 23.3–25.3 |
| High SDI        | 24.3      | 23.3–25.3 |

# Discussion and Limitations

## Discussion

- SDI gradient likely reflects diagnostic/certification capacity (“diagnostic divide”).
- Low  $R^2$  suggests major roles for age structure, health systems, and coding variation.
- Post-2020 APC rise aligns with COVID-era vulnerability and care disruption.
- Hungary’s mid-range burden supports regional preparedness and continuity of care.

## Limitations

- Ecological country-level analysis → no individual-level causality.
- GBD secondary estimates depend on national coding/data quality.
- $R^2 = 0.0577$  indicates substantial unexplained variability.



# Conclusion

- 1 SDI is significantly associated with recorded Alzheimer's mortality ( $p < 0.001$ ), but  $R^2 = 0.0577$  explains little variance—SDI is not the whole story.
- 2 Dunn's post-hoc shows all SDI group pairs differ except Low-middle vs Middle (SDI2 vs SDI3) ( $p = 0.2875$ ).
- 3 Joinpoint shows post-2020 acceleration (APC 2020–2023 = 0.91%, significant).
- 4 Hungary is mid-range in Central Europe in 2023 (ASMR 21.1287) with a modest post-2020 increase (+1.95%).
- 5 Global 2030 forecast: 25.8 per 100,000 (Joinpoint/APC-based 95% CI: 24.2–27.4); SDI-group forecasts mortality rate per 100,000 (20.4–27.1) with 95% prediction intervals.

# Appendix: My Contribution

- Data Extraction from the GBD 2023 database
- Data Cleaning
- Conducted the Linear Regression, Kruskal-Wallis, and Joinpoint analyses using Stata and Joinpoint Regression Software.
- I created all the visualizations presented.

Ethical Statement: The study utilizes publicly available aggregate data; no separate ethical approval was required.

# THANK YOU

Supervised by :

- Dr. Toth Agnes
- Dr. Ghanem Amr Sayed



## References(Key):

- IHME. Global Burden of Disease (GBD) 2023 Results Tool.
- National Cancer Institute. Joinpoint Regression Program.
- WHO. Dementia: key facts.

